Lower back pain assessment

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This tool supports decisions about how soon to seek care and which specialist to see. It does not diagnose or treat, and it does not replace an examination by a doctor. If you feel very unwell or are worried, go to a hospital.

See a doctor within 24–48 hours

Do not wait for it to settle. Book the earliest appointment or attend a hospital OPD.

- Unable to walk or stand
- Leg pain with nerve-root features
- Strong pain (7–8 out of 10)

CLINICAL IMPRESSION

Most likely: Lumbar disc herniation with nerve root irritation (sciatica) [Strong match]

Lumbar disc herniation · lumbosacral radiculopathy

Supports: Symptoms below the knee · Straight leg raise positive · Leg pain is worse · Sharp, electric or shooting pain down the leg · Pain spreads into the leg · Sitting · Bending forward

A disc between two vertebrae bulges or herniates backwards and presses a nerve root. Pain runs into the buttock and leg along that root, often worse when sitting, bending or coughing.

Why your answers point here

- Symptoms below the knee — Symptoms below the knee point to a nerve root rather than a muscle, because muscle pain rarely travels that far.
- Straight leg raise positive — A positive straight leg raise stretches the lower nerve roots; pain past the knee means a root is irritated (92% sensitive).
- Leg pain is worse — Leg pain worse than back pain is a classic feature of sciatica.
- Sharp, electric or shooting pain down the leg — Sharp, electric pain running along a line is typical of an irritated nerve root.
- Pain spreads into the leg — Pain spreading from the back into the leg shows that a nerve or its root may be involved.
- Sitting — Sitting raises pressure inside the lumbar discs, so disc problems hurt more when you sit.
- Bending forward — Bending forward squeezes the front of the disc and pushes its gel backwards towards the nerves.

Why it happens: Discs slowly lose water and height from the late 20s. Long sitting, repeated bending, lifting with a rounded back, smoking and a desk job load the back of the disc until a crack in the outer ring lets the gel push out. The final trigger is often small: getting up, bending to wear shoes, a sneeze.

How doctors confirm it: Neurological examination (straight leg raise, strength, reflexes, sensation). MRI of the lumbar spine if there is weakness, a red flag, or no improvement after about 6 weeks. Plain X-rays do not show discs.

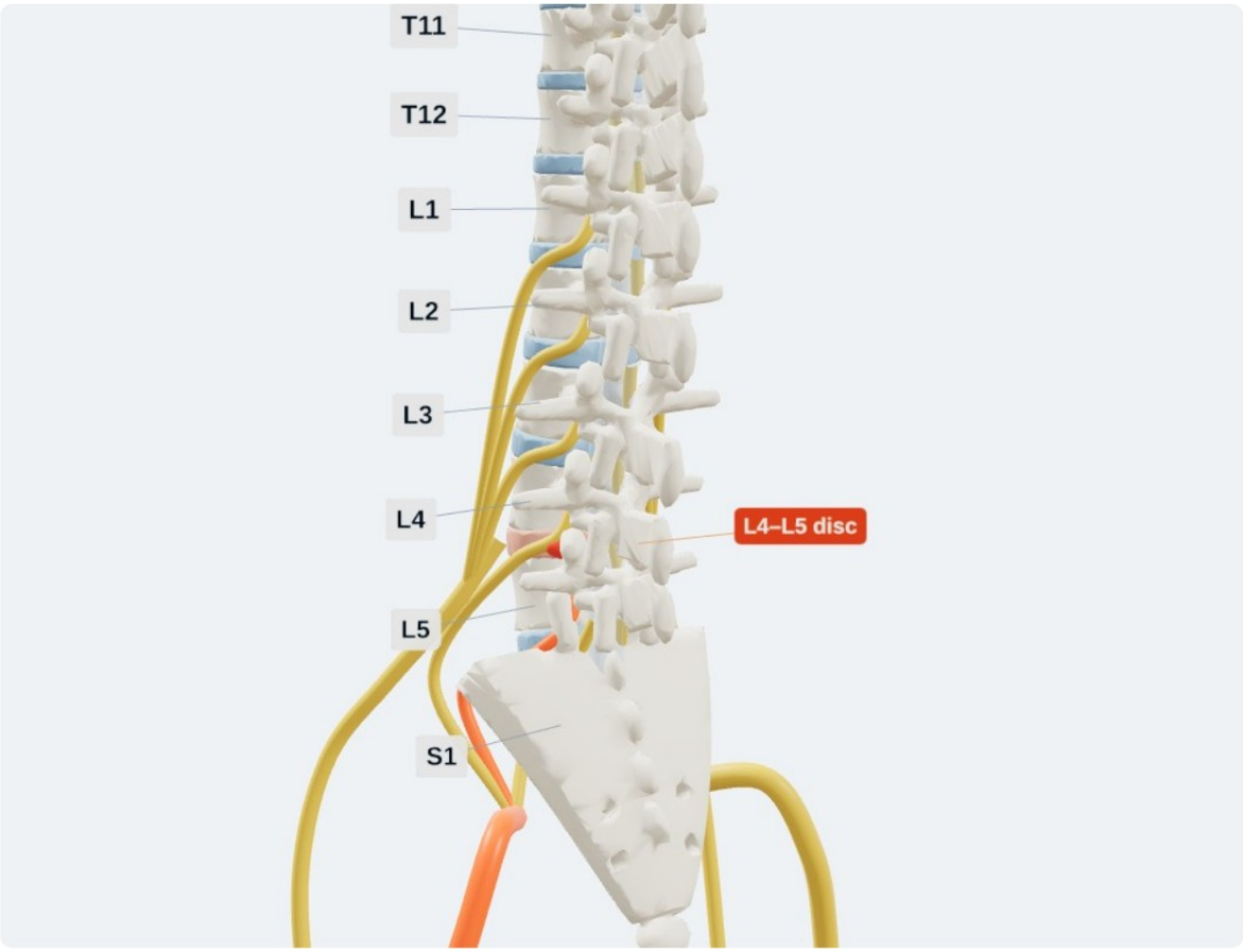
How it is treated: First 6 weeks: stay active, a short course of pain medicine chosen by your doctor, and physiotherapy (nerve gliding, core and hip strengthening). If leg pain persists: an image-guided epidural steroid injection. Surgery (microdiscectomy) for worsening weakness, cauda equina, or severe pain beyond 6–12 weeks.

Outlook: About 90% of people improve within 3 months without surgery. Larger herniations shrink on their own more often: extrusions in about 70% and sequestrations in about 96%. Surgery (microdiscectomy) is considered for weakness or severe pain that does not settle.

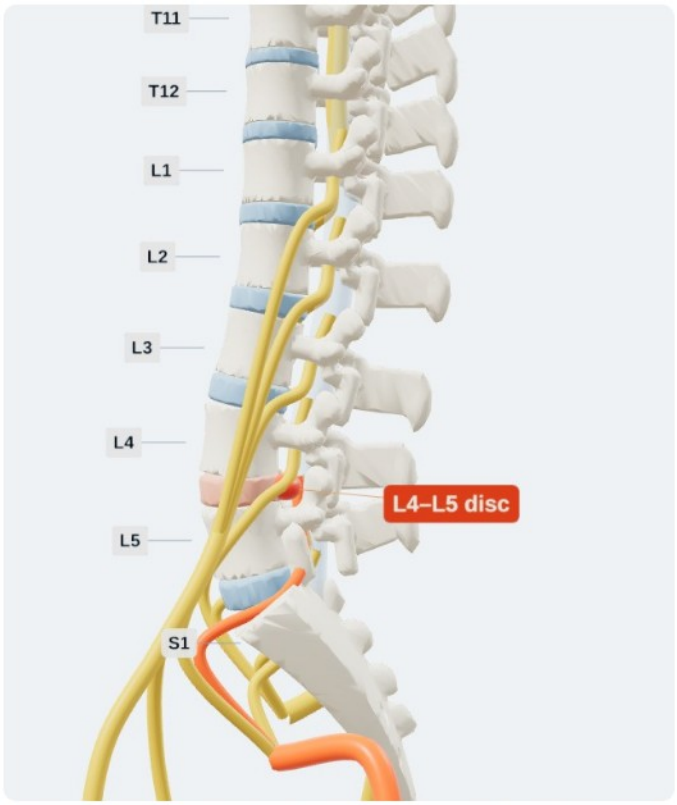
A DOCTOR'S EXAMINATION IS ESSENTIAL

- This assessment cannot test your reflexes, muscle strength or skin sensation, which show which nerve is involved and how badly.
- Only a doctor can decide whether you need an MRI, X-ray or blood tests, and interpret them alongside your examination.
- Similar pain can come from the hip, kidney, pelvis or abdomen; a doctor examines these too.
- Medicines, injections and exercise plans must be chosen for you personally.

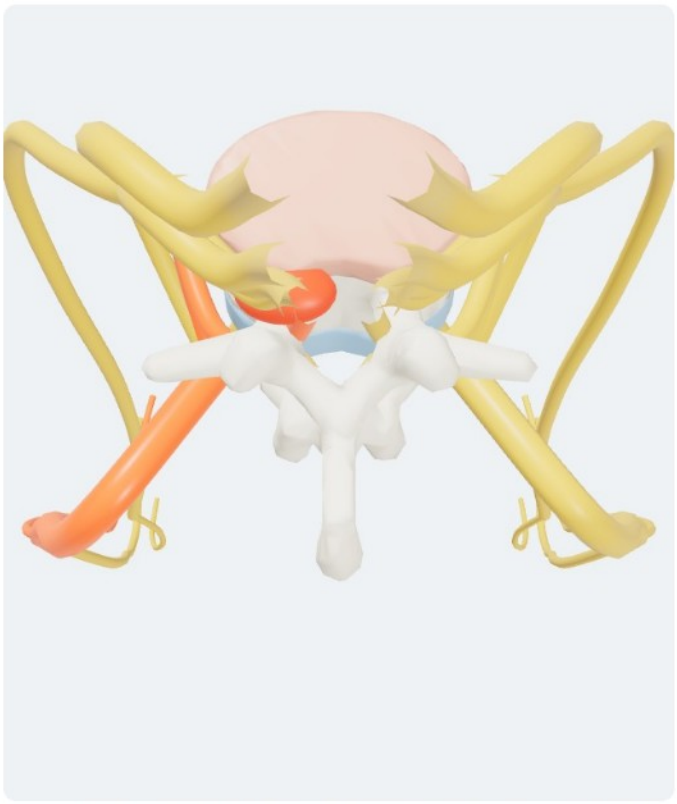
3D ANATOMY OF YOUR PROBLEM



Lumbar spine from behind, vertebrae numbered. Pink: affected disc. Red: the herniated part. Orange: the pressed nerve root and its path into the leg.



Side view: the disc bulging backwards into the space where the nerve roots run.



Looking down on the L4–L5 disc with the vertebra above removed, as on an MRI cross-section.



Nerve path: the affected root continues into the leg, which is why pain travels where you marked it.



Areas you marked, back view



Areas you marked, front view

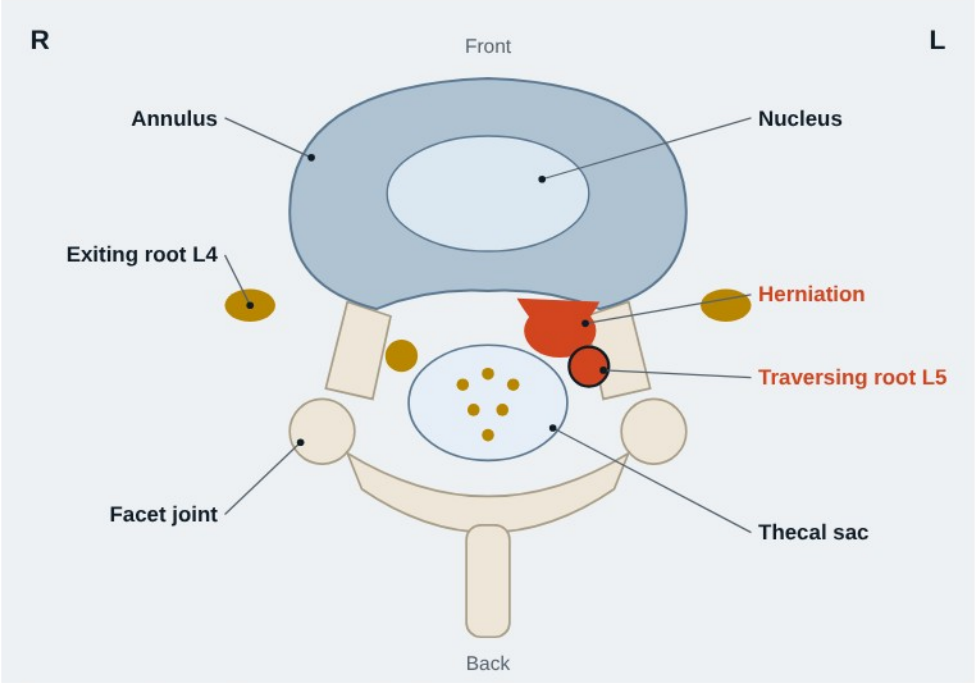
ROOT CAUSE EXPLAINED

Your answers fit a disc at L4–L5 pressing on the L5 nerve root on the left side. Each disc has a tough outer ring (annulus fibrosus) around a gel centre (nucleus pulposus). A sudden load such as getting up quickly, bending or lifting can tear the ring, and the gel pushes backwards into the space where the nerve roots run.

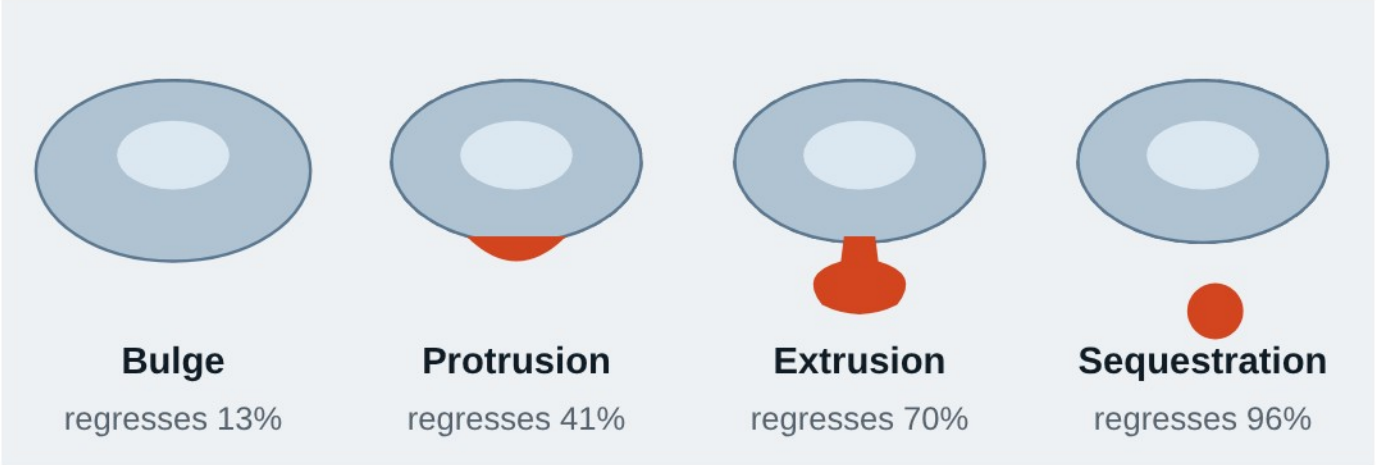
The pressed root becomes inflamed. That is why pain travels along its path into the buttock and leg (sciatica), and why sitting, bending and coughing, which raise pressure inside the disc, make it worse.

About 90–95% of herniations sit just off-centre (paracentral) and press the root travelling down past the disc, so an L4–L5 disc usually affects the L5 root. The less common foraminal type (5–10%) presses the root leaving at that level.

Only an MRI shows the exact level and type. MRI also finds bulges in many people without pain (30% at age 20, 84% at 80), so findings are always matched to symptoms.



Cross-section at the disc, viewed from the feet as on an MRI (your left is on the right).



NERVE ROOT REFERENCE

L5 — Pain and numbness: Shin and top of the foot; pain in the buttock, back-outer thigh and outer leg. Weakness: Big-toe extension, ankle dorsiflexion (foot drop). Reflex: None reliable. Usual disc: L4–L5.

OTHER POSSIBILITIES

• Piriformis syndrome [Possible] — Buttock, Back of the thigh, Sitting

WHICH SPECIALIST TO SEE

1. Orthopaedics – Spine (Start here)
Bone, joint and spine specialist; first stop for most back and sciatica problems. For: Lumbar disc herniation with nerve root irritation (sciatica).

2. Neurosurgery / Spine surgery
For nerve compression with weakness, or pain that does not settle with treatment. For: Lumbar disc herniation with nerve root irritation (sciatica).

3. Physiotherapy / PM&R
Structured exercise and rehabilitation; the core of treatment for most back pain. For: Lumbar disc herniation with nerve root irritation (sciatica).

YOUR ANSWERS

Age	30–39 · Male
Medical history (tick any that apply)	None
Areas marked	Lower back, centre; Lower back, beside the spine (left); Buttock (left); Back of the thigh (left) [S1–S2]; Outer shin, top of the foot, big toe (left) [L5]
Which troubles you more: back pain or leg pain?	Leg pain is worse
How did it start?	Getting up quickly from bed or a chair
When did it start?	This bout started within the last week
How does the pain behave?	Sharp, electric or shooting pain down the leg; Sitting; Bending forward
Impact on daily life	Pain right now 8/10 · Walking: Cannot walk or stand · Sitting: Cannot sit · Sleep: —
Straight leg raise (SLR)	Positive
Crossed straight leg raise	Not answered
Heel walk (L4–L5 strength)	Not answered
Single-leg heel raise (S1 strength)	Not answered
Red-flag screen	No to all warning signs

SOURCES

1. WHO guideline on chronic low back pain (summary, Health Policy Watch, 2023) — https://healthpolicy-watch.news/who-issues-first-guidelines-low-back-pain/
2. Orthobullets – Lumbar disc herniation — https://www.orthobullets.com/spine/2035/lumbar-disc-herniation
3. Chiu CC et al. Probability of spontaneous regression of lumbar herniated disc. Clin Rehabil 2015;29:184–95 (summary) — https://www.painscience.com/biblio/intervertebral-disc-injuries-spontaneously-regress-surprisingly-often--and-worse-ones-more.html
4. Cochrane review CD007431 – Physical examination for lumbar radiculopathy due to disc herniation — https://www.cochrane.org/evidence/CD007431_physical-examination-diagnosis-lumbar-radiculopathy-due-disc-herniation-patients-low-back-pain-and

5. Brinjikji W et al. Imaging features of spinal degeneration in asymptomatic populations. AJNR 2015;36:811–6 — https://www.ajnr.org/content/36/4/811
6. Radiology Assistant – Lumbar disc nomenclature 2.0 — https://radiologyassistant.nl/neuroradiology/spine/lumbar-disc-nomenclature-2-0
7. North American Spine Society – Bulge vs herniation — https://www.spine.org/KnowYourBack/Conditions/Degenerative-Conditions/Bulge-Herniation
8. MSD Manual – Symptoms of common radiculopathies by dermatomal level — https://www.msdmanuals.com/professional/multimedia/table/symptoms-of-common-radiculopathies-by-dermatomal-level
9. Physiopedia – Lumbar radiculopathy assessment (myotomes, reflexes) — https://www.physio-pedia.com/Lumbar_Radiculopathy_Assessment
10. Medmastery – Diagnosing lumbar radiculopathy — https://www.medmastery.com/guide/neurology-diseases-clinical-guide/diagnosing-lumbar-radiculopathy
11. NICE NG59 – Low back pain and sciatica in over 16s — https://www.nice.org.uk/guidance/ng59/chapter/Recommendation
12. Royal Free London NHS – Cauda equina syndrome warning signs — https://royalfree.nhs.uk/patients-and-visitors/patient-information-leaflets/cauda-equina-syndrome
13. WikEM – Back pain red flags — https://wikem.org/wiki/Back_pain_(red_flags)
14. MSD Manual – Lumbar spinal stenosis — https://www.msdmanuals.com/professional/musculoskeletal-and-connective-tissue-disorders/neck-and-back-pain/lumbar-spinal-stenosis
15. ASAS criteria for inflammatory back pain — https://empendium.com/mcmtextbook/table/031_6932
16. Spine-health – Vertebral fracture symptoms — https://spine-health.com/conditions/osteoporosis/vertebral-fracture-symptoms
17. Pott's disease (spinal tuberculosis) — https://en.wikipedia.org/wiki/Pott%27s_disease
18. NHS 111 Wales – Kidney stones — https://111.wales.nhs.uk/kidneystones/
19. Texas Orthopedics – Coccydynia — https://www.txortho.com/patient-education-article/coccydynia-tailbone-pain-969
20. Medical emergency numbers in India (108, 112) — https://zhl.org.in/blog/medical-emergency-numbers-india/
21. GBD 2021 Neck Pain Collaborators. Global burden of neck pain 1990–2020 and projections to 2050. Lancet Rheumatol 2024;6:e142–55 — https://research.uaeu.ac.ae/en/publications/global-regional-and-national-burden-of-neck-pain-19902020-and-pro/
22. Radhakrishnan K et al. Epidemiology of cervical radiculopathy, Rochester 1976–1990. Brain 1994 (PubMed abstract) — https://pubmed.ncbi.nlm.nih.gov/8186959/
23. WikEM – Cervical radiculopathy (exam by level) — https://wikem.org/wiki/Cervical_Radiculopathy
24. Physiopedia – Clinical prediction rule for cervical radiculopathy (Wainner 2003) — https://physio-pedia.com/CPR_for_Cervical_Radiculopathy
25. Orthobullets – Cervical myelopathy — https://www.orthobullets.com/spine/2031/cervical-myelopathy
26. UK Health Security Agency – Meningitis: don't ignore the signs — https://gov.uk/government/publications/meningitis-signs-and-symptoms-poster/meningitis-dont-ignore-the-signs-and-symptoms
27. Heart attack warning signs (American Heart Association) — https://oaklawnhospital.org/services/cardiology/warning-signs-of-a-heart-attack-american-heart-association/
28. Healthdirect (Australian Government) – Upper back (thoracic) pain — https://www.healthdirect.gov.au/upper-back-thoracic-pain
29. Patient.info – Shingles (herpes zoster) — https://patient.info/health/Shingles.htm
30. Wolfe F et al. 2016 revisions to the fibromyalgia diagnostic criteria (criteria form) — https://lin-web.clarkson.edu/~lrussek/2016FMS.pdf